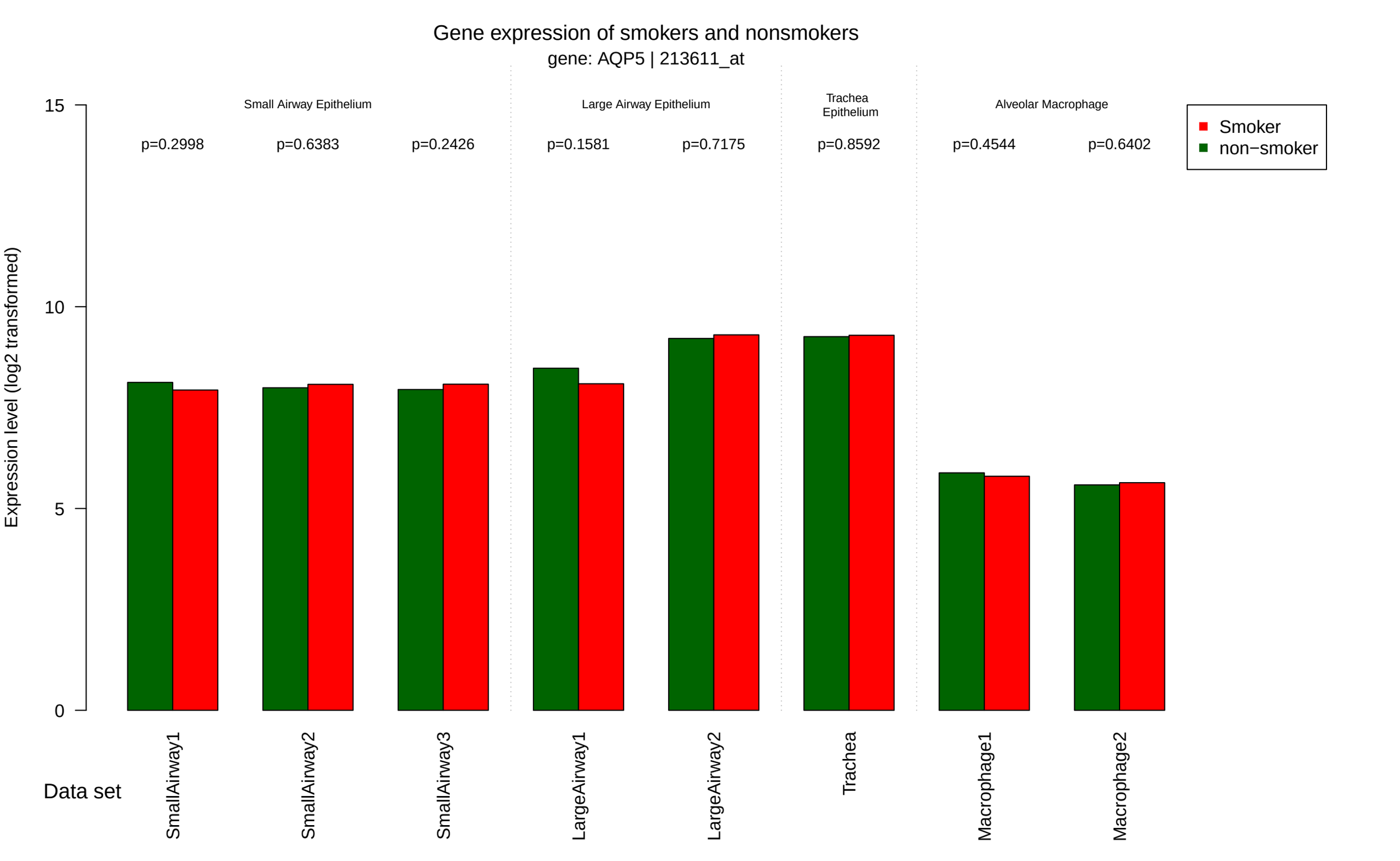
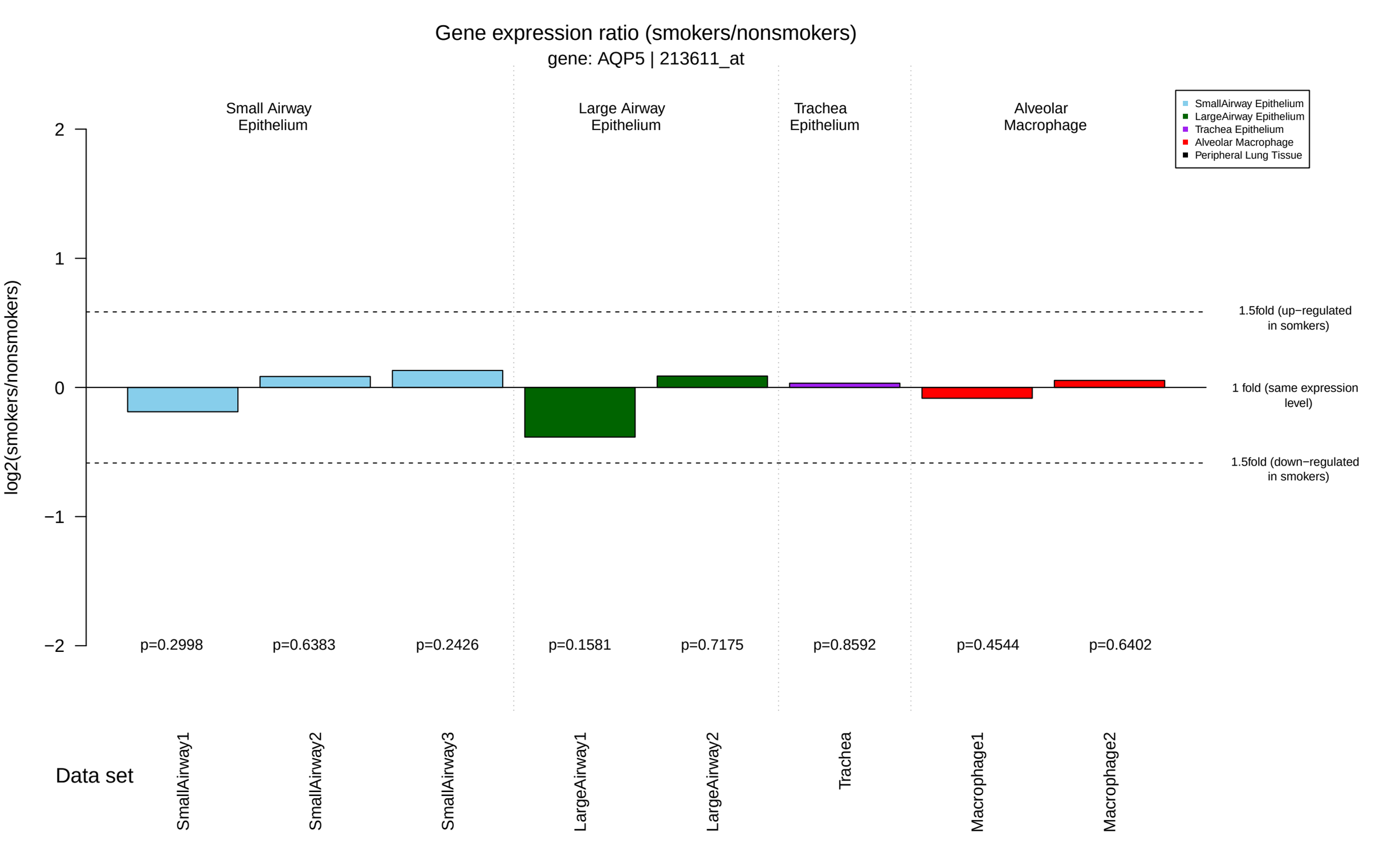
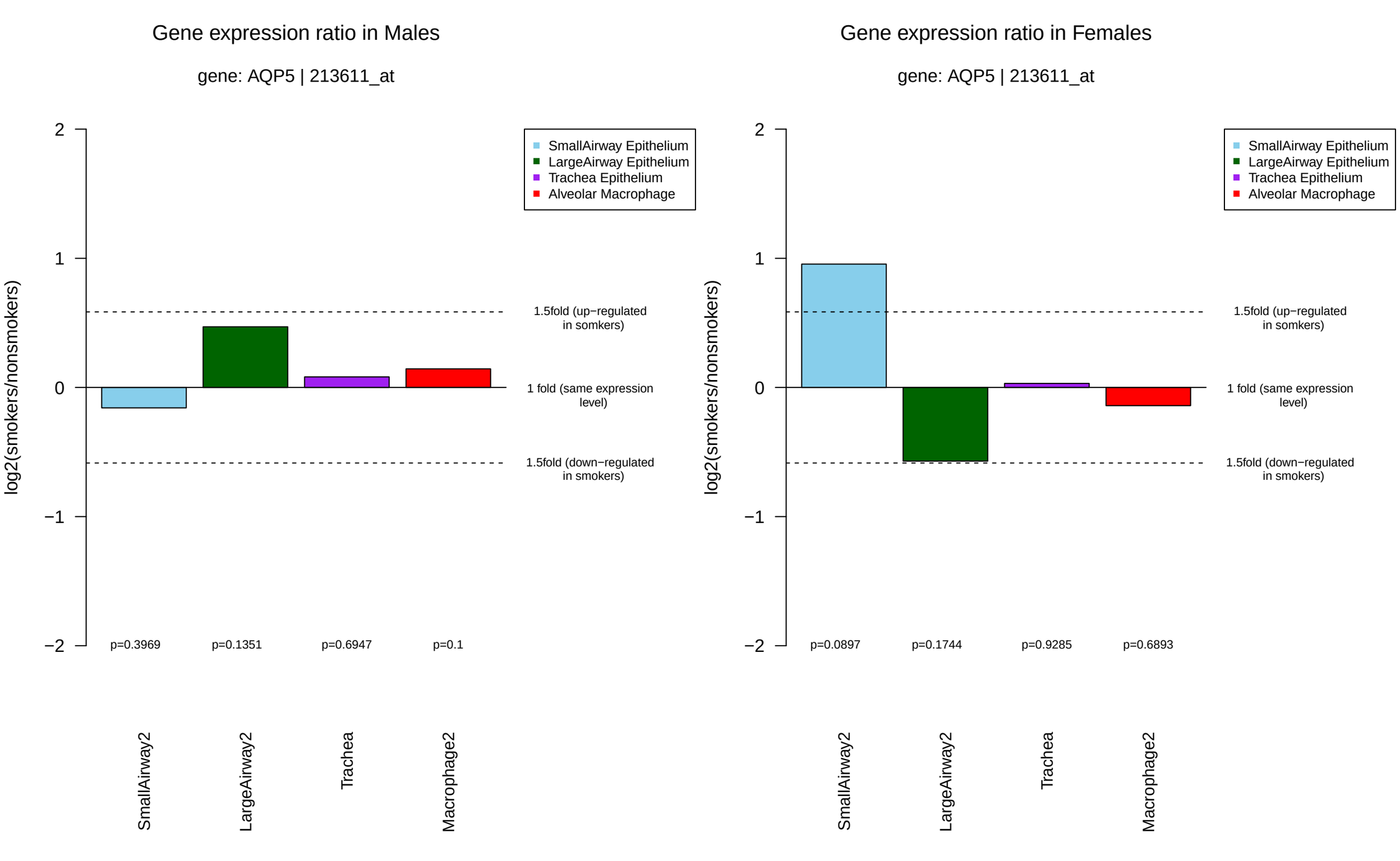
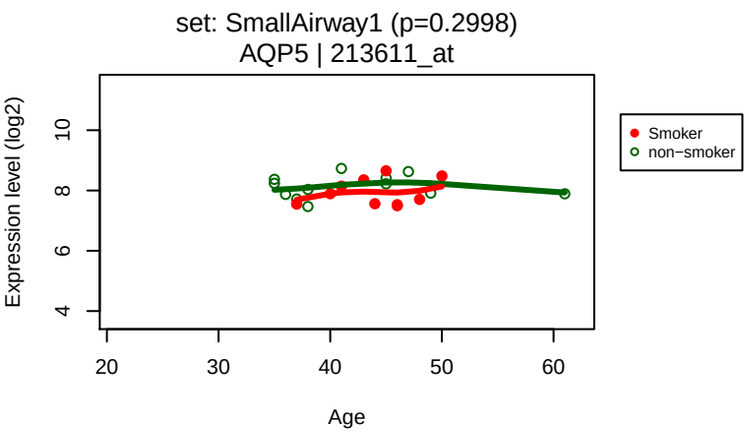
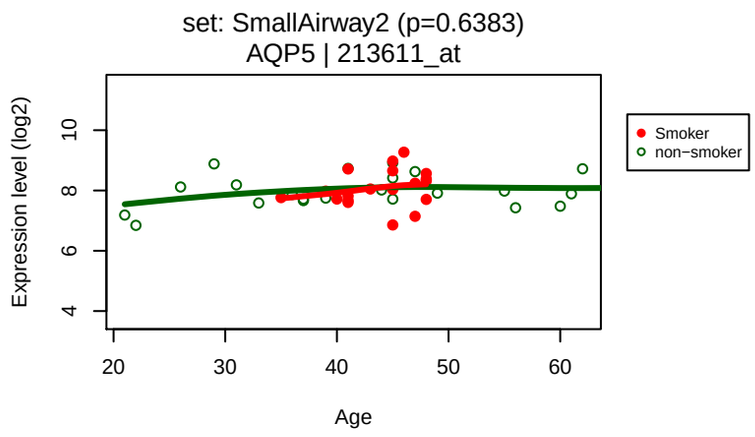
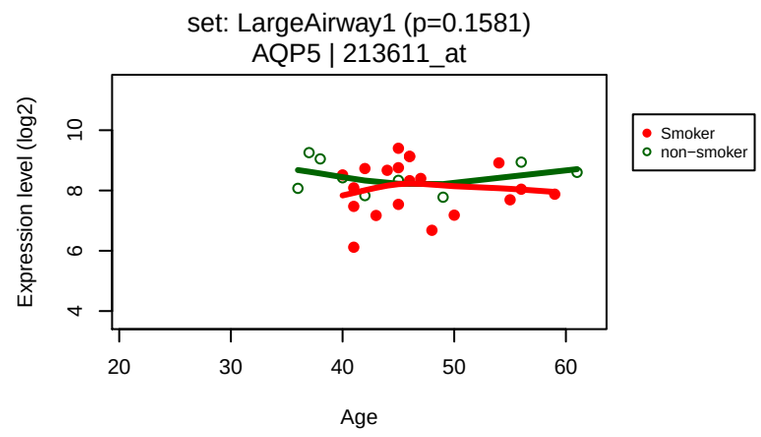

Gene expression of smokers and nonsmokers
gene: AQP5 | 213611_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2998
p=0.6383
p=0.2426
p=0.1581
p=0.7175
p=0.8592
p=0.4544
p=0.6402
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

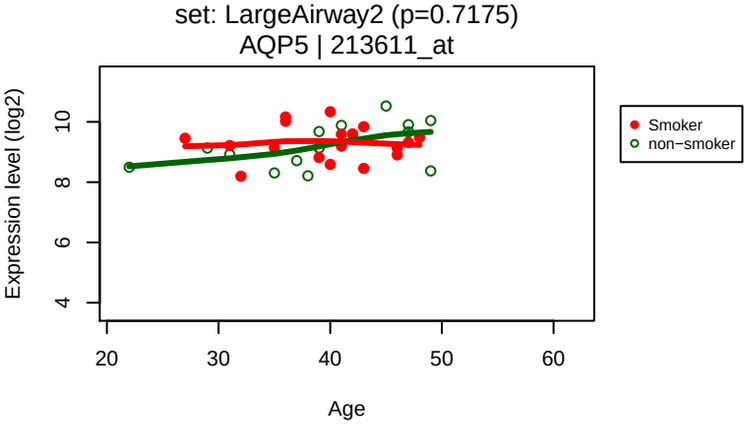

Gene expression ratio (smokers/nonsmokers)
gene: AQP5 | 213611_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2998
p=0.6383
p=0.2426
p=0.1581
p=0.7175
p=0.8592
p=0.4544
p=0.6402
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

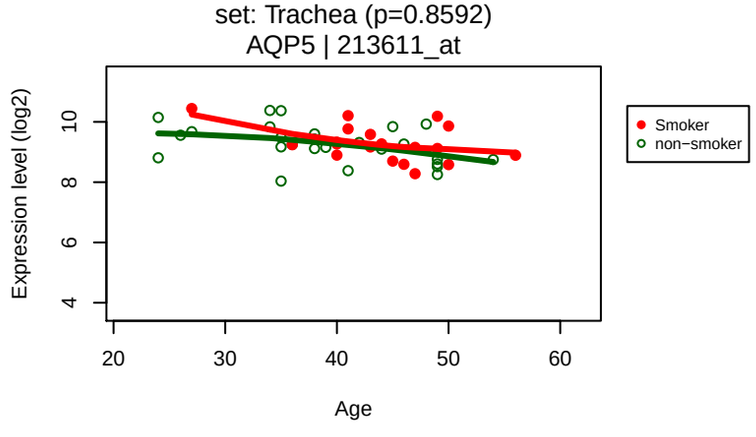

Gene expression ratio in Males
gene: AQP5 | 213611_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3969
p=0.1351
p=0.6947
p=0.1
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP5 | 213611_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0897
p=0.1744
p=0.9285
p=0.6893
SmallAirway2
LargeAirway2
Trachea
Macrophage2

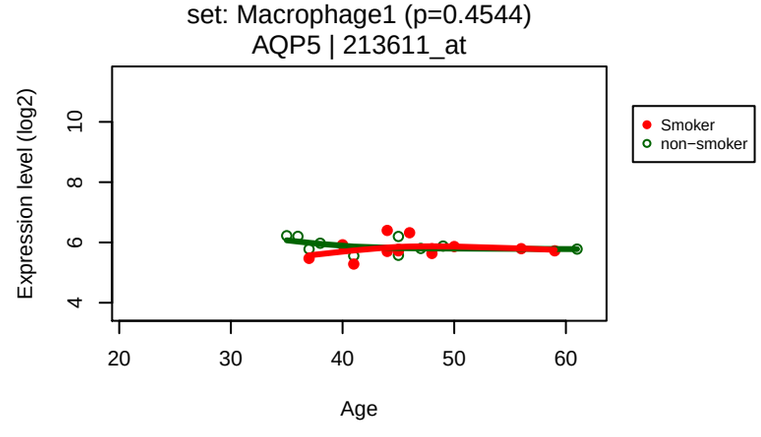

set: SmallAirway1 (p=0.2998)
AQP5 | 213611_at
Expression level (log2)
Age
Smoker
non−smoker

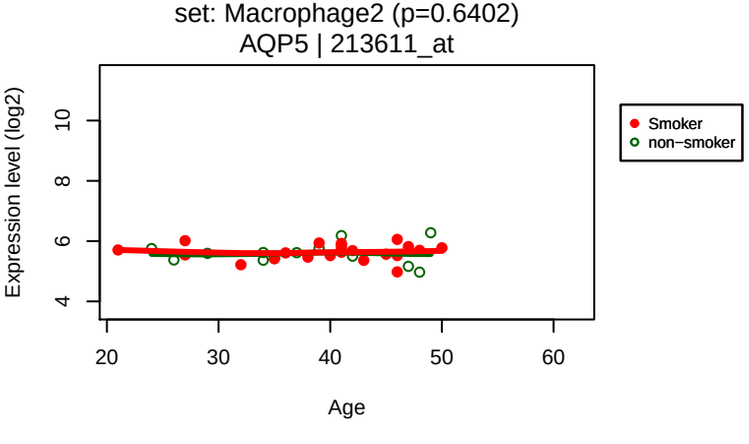

set: SmallAirway2 (p=0.6383)
AQP5 | 213611_at
Expression level (log2)
Age
Smoker
non−smoker

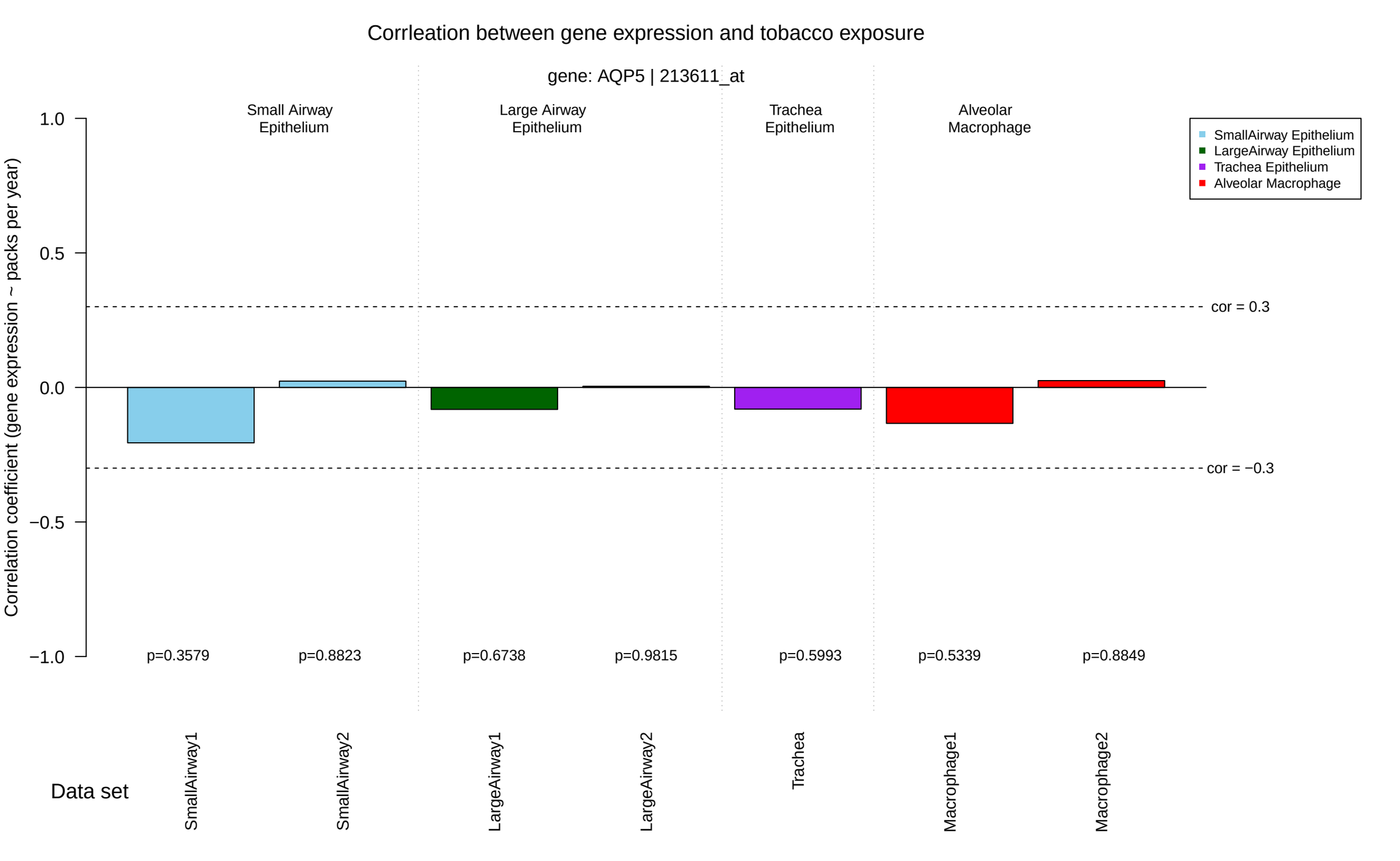

set: LargeAirway1 (p=0.1581)
AQP5 | 213611_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7175)
AQP5 | 213611_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8592)
AQP5 | 213611_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4544)
AQP5 | 213611_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6402)
AQP5 | 213611_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP5 | 213611_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3579
p=0.8823
p=0.6738
p=0.9815
p=0.5993
p=0.5339
p=0.8849
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2